BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Waterfall.

Q2. 10.

Q3. A link/reference to the next node (for a singly linked list).

Q4. In its left subtree (under the chosen no-duplicate convention).

You Do pages · Independent questions

Q1. Test first/red-green-refactor cycle [3]; two developers collaborating with shared review/knowledge [3].

Q2. Visible working prototypes [2], frequent user feedback [2], rapid refinement of requirements [2].

Q3. XP testing/CI benefits [2], RAD feedback benefits [2], rushed assurance/traceability risks [2], justified safeguards/recommendation [2].

Exit Ticket · Model answer

False. It improves structure while preserving observable behaviour.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 118 · Lesson 118 · Extreme Programming and RAD

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which lifecycle uses relatively fixed sequential stages?

Q2. For the binary value 1010, what is its decimal value?

Q3. A linked-list node has data and what other essential field?

Q4. In a BST, where is a value less than the current node placed?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

XP — Agile method emphasising disciplined engineering practices.

TDD — Write a failing test then implement/refactor.

RAD — Development centred on rapid prototyping and user feedback.

refactoring — Improving structure without changing observable behaviour.

New learning

XP practices — Test-driven development, pair programming, refactoring and continuous integration.

RAD approach — Rapid application development uses prototyping and stakeholder collaboration.

Balance cost and fit — Consider user availability, architecture, maintainability and regulatory oversight.

Worked example

A small team repeatedly breaks existing features after changes. Which XP practices help?

1. Model answer — Automated tests written before/with code (TDD) define intended behaviour, continuous integration runs regression tests on each change, and refactoring improves design without changing observable behaviour. — Check: link each claim to the scenario and use precise OCR terminology.

We Do · Practise with guidance

A local club cannot describe its booking interface but can test clickable prototypes. Explain RAD.

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Create an early working prototype, invite feedback, revise the interface rapidly and iteratively converge on requirements. Beware treating a rushed prototype as production-quality architecture without further work.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain test-driven development and pair programming within XP. [6]

Question 2

Describe two reasons RAD may suit a system with unclear interface requirements. [6]

Question 3

Evaluate XP and RAD for building a safety-critical medication scheduler under time pressure. [8]

Exit Ticket · Answer before checking

Refactoring is intentionally changing the program's externally visible requirements.